

Image Enhancement in the Frequency Domain

Subtopic 3.4: Sharpening Frequency Domain Filters

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Outline - Subtopic 3.4: Sharpening Frequency Domain Filters

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High-Pass Filters: Overview

- **Opposite of Low-Pass:** Allow high frequencies to pass, attenuate low frequencies
- **Purpose of Sharpening:**
 - Enhance edges and fine details
 - Increase image contrast
 - Emphasize rapid intensity changes
 - Remove blurring effects
- **Complementary Relationship:**

$$H_{\text{High-Pass}}(u, v) = 1 - H_{\text{Low-Pass}}(u, v) \quad (1)$$

- **Key Characteristics:**
 - DC component (average value) is attenuated or removed
 - Edge details are enhanced
 - May amplify noise

High-Pass Filter Types

- **Four Main Types:**

- ① Ideal High-Pass Filter (IHPF)
- ② Butterworth High-Pass Filter (BHPF)
- ③ Gaussian High-Pass Filter (GHPF)
- ④ Laplacian Filter

- **Differences from Low-Pass:**

- Inverse frequency relationship
- Enhanced high-frequency components
- Potential noise amplification
- Often combined with low-pass for band-pass filtering

- **Applications:**

- Edge detection and enhancement
- Image sharpening and deblurring
- Feature extraction
- Detail preservation in preprocessing

Ideal High-Pass Filter: Definition

- **Mathematical Definition:**

$$H(u, v) = \begin{cases} 0 & \text{if } D(u, v) \leq D_0 \\ 1 & \text{if } D(u, v) > D_0 \end{cases} \quad (2)$$

where $D(u, v)$ is distance from origin in frequency domain

- **Complementary to ILPF:**

$$H_{\text{HPF}}(u, v) = 1 - H_{\text{LPF}}(u, v) \quad (3)$$

- **Distance Function:**

$$D(u, v) = \sqrt{(u - M/2)^2 + (v - N/2)^2} \quad (4)$$

- **Cutoff Frequency:** D_0 determines transition point

IHPF: Characteristics and Properties

- **Key Features:**

- Blocks all frequencies within radius D_0 (sets to zero)
- Passes all frequencies beyond D_0 (unattenuated)
- Sharp discontinuity at cutoff

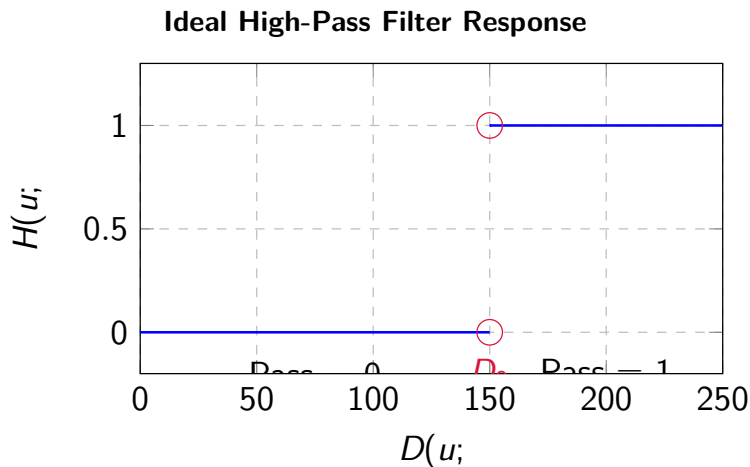
- **Advantages:**

- Simple to understand and implement
- Sharp frequency cutoff
- Maximum edge enhancement at cutoff

- **Disadvantages:**

- **Severe Ringing:** Oscillations around edges (Gibbs phenomenon)
- Removes too much low-frequency information
- Creates image with reduced brightness
- Poor practical performance
- Discontinuous response causes artifacts

IHPF: Visual Representation



Blocks low frequencies; passes high frequencies

IHPF: Problem with DC Removal

- **DC Component Effect:**

- At origin $(0, 0)$: $D(0, 0) = 0 < D_0 \Rightarrow H(0, 0) = 0$
- DC component (average brightness) is removed
- Results in dark or grayish image
- Loss of mean intensity information

- **Mathematical Consequence:**

$$\sum_{x,y} g(x, y) = 0 \quad (\text{zero mean output}) \quad (5)$$

- **Solution:** Add back partial DC component

$$H_{\text{modified}}(u, v) = a + (1 - a) \cdot H(u, v), \quad 0 < a < 1 \quad (6)$$

Butterworth High-Pass Filter: Definition

- **Mathematical Definition:**

$$H(u, v) = \frac{1}{1 + \left[\frac{D_0}{D(u, v)} \right]^{2n}} \quad (7)$$

where $D(u, v)$ is distance and n is filter order

- **Relationship to BLPF:**

$$H_{\text{BHPF}}(u, v) = 1 - H_{\text{BLPF}}(u, v) \quad (8)$$

- **At Cutoff Frequency ($D = D_0$):**

$$H(D_0, v) = \frac{1}{1 + 1} = 0.5 \quad (-3 \text{ dB point}) \quad (9)$$

- **Key Parameter n :** Controls transition sharpness

BHPF: Order Effect and Properties

- **Order Behavior:**

- $n = 1$: Smooth transition, gradual response
- $n = 2$: Moderate transition
- $n = 4$: Sharper transition
- $n \rightarrow \infty$: Approaches ideal high-pass filter

- **Transition Characteristics:**

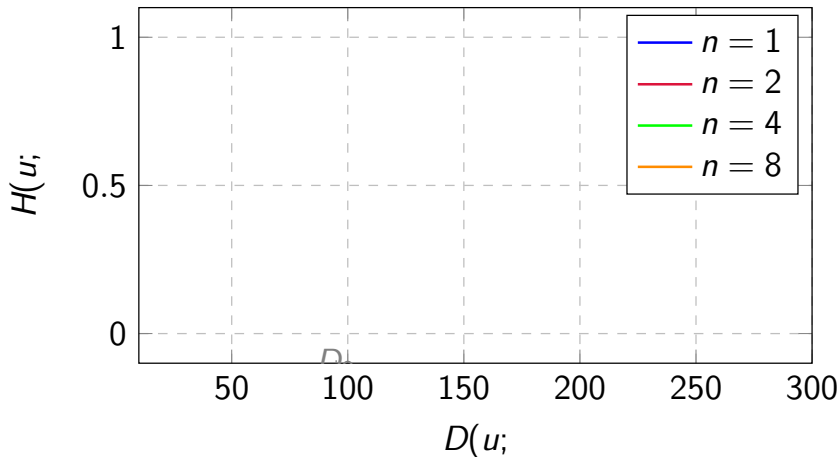
- Monotonically increasing response
- Smooth, continuous function
- Reduced ringing compared to IHPF
- No sharp discontinuity

- **DC Component Behavior:**

- At origin: $H(0, 0) = \frac{1}{1+\infty} \approx 0$ (DC removed)
- Some low-frequency information retained
- Better than IHPF but DC still attenuated

BHPF: Visual Comparison for Different Orders

Butterworth High-Pass Filter for Various Orders



Higher order approaches ideal high-pass behavior

BHPF: Advantages and Applications

- **Advantages:**

- Smooth transition without sharp discontinuity
- Adjustable sharpness with order parameter
- Less ringing than IHPF
- Better control over edge enhancement
- Practical implementation

- **Trade-offs:**

- Higher order $\Rightarrow$ sharper cutoff but more computation
- Lower order $\Rightarrow$ smoother but less selective
- Order $n = 2$ often provides good balance

- **Typical Applications:**

- Image sharpening and edge enhancement
- Detail preservation in preprocessing
- Feature extraction

Gaussian High-Pass Filter: Definition

- **Mathematical Definition:**

$$H(u, v) = 1 - e^{-D^2(u, v)/(2D_0^2)} \quad (10)$$

where $D(u, v)$ is distance from origin, D_0 is cutoff frequency

- **Complementary to GLPF:**

$$H_{\text{GHPF}}(u, v) = 1 - H_{\text{GLPF}}(u, v) \quad (11)$$

- **Alternative Form:**

$$H(u, v) = 1 - e^{-\sigma^2 D^2(u, v)/2} \quad (12)$$

where σ controls the width of transition

- **At Cutoff:**

$$H(D_0, D_0) = 1 - e^{-1/2} \approx 0.3935 \quad (13)$$

GHPF: Properties and Characteristics

- **Key Properties:**

- **Smooth:** No sharp cutoff, smooth Gaussian curve
- **No Ringing:** Spatial domain has no oscillations
- **Self-Dual:** Gaussian in both frequency and spatial domains
- **Minimal Artifacts:** Very smooth transitions

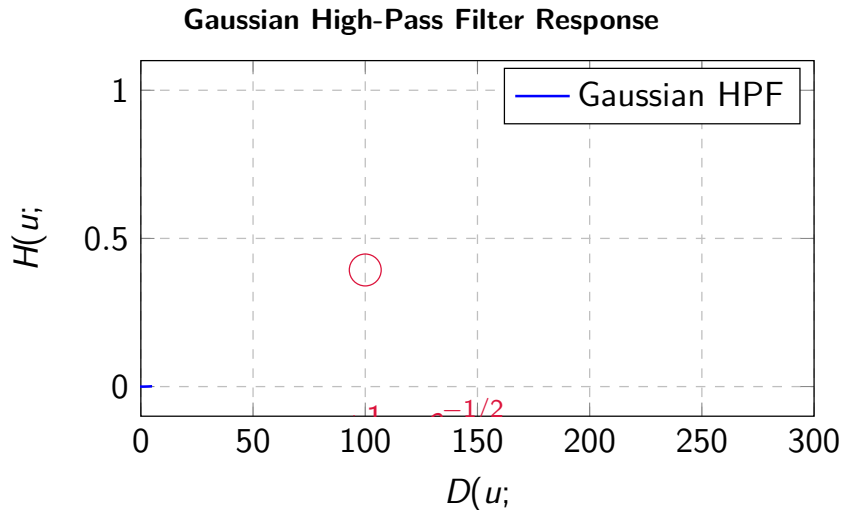
- **DC Component Behavior:**

- At origin: $H(0, 0) = 1 - e^0 = 0$ (DC completely removed)
- Smoother removal than BHPF
- Better for emphasizing details only

- **Comparison with BHPF:**

- Smoother response curve
- Better artifact reduction
- Preferred in practice

GHPF: Visual Representation



Smooth, inverse Gaussian response with no sharp cutoff

GHPF: Advantages and Applications

- **Advantages:**

- **No Ringing:** Smoothest possible transition
- **Best Quality:** Most natural-looking edge enhancement
- **Mathematical Elegance:** Self-dual property
- **Easy to Use:** Simple parameter (D_0)
- **Stable:** Numerically stable computation

- **Applications:**

- Image sharpening with minimal artifacts
- Edge detection and enhancement
- Detail preservation in preprocessing
- Medical image analysis
- High-quality image enhancement

- **Practical Preference:** Often preferred over BHPF

Laplacian Filter: Overview

- **Definition:** Second derivative operator for edge detection
- **Spatial Domain Laplacian:**

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} \quad (14)$$

- **Frequency Domain Equivalence:**
 - Multiplication by $-D^2(u, v)$ in frequency domain
 - Emphasizes rapid changes (high frequencies)
 - Strong edge detection capability
- **Key Characteristic:**
 - Even stronger high-pass emphasis than Gaussian HPF
 - Magnitude increases with distance from origin
 - Ideal for detecting discontinuities

Laplacian: Mathematical Definition

- **Frequency Domain Representation:**

$$H(u, v) = -4\pi^2 D^2(u, v) = -4\pi^2 [(u - M/2)^2 + (v - N/2)^2] \quad (15)$$

- **Normalized Form (for magnitude):**

$$H(u, v) = -D^2(u, v) \quad (16)$$

- **Spatial Domain Equivalent:**

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix} \quad (17)$$

Laplacian: Characteristics

- **Key Properties:**

- **Isotropic:** Same response in all directions
- **Second-Order Derivative:** More sensitive to detail changes
- **Zero at DC:** Removes average intensity
- **Increases with Distance:** Response grows quadratically with frequency

- **Advantages:**

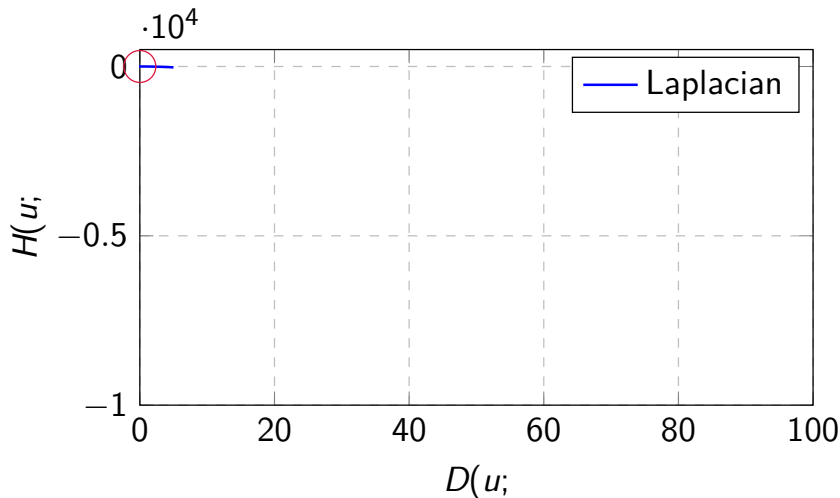
- Excellent edge detection
- Detects discontinuities of any orientation
- Emphasizes fine details
- Mathematically elegant

- **Disadvantages:**

- Very sensitive to noise
- Removes low-frequency information completely
- Output image has zero mean
- Requires combination with original for visualization

Laplacian: Visual Representation

Laplacian Filter Response (2-D Profile)



Laplacian: Image Enhancement

- **Problem:** Pure Laplacian output has negative values and zero mean
- **Solution 1: Combination with Original**

$$g(x, y) = f(x, y) - c \cdot \nabla^2 f(x, y) \quad (18)$$

where c is a weighting constant

- **Solution 2: Frequency Domain Combination**

$$H_{\text{combined}}(u, v) = 1 + c \cdot H_{\text{Laplacian}}(u, v) \quad (19)$$

- **Solution 3: Combine with Butterworth or Gaussian**

$$H(u, v) = H_{\text{HPF}}(u, v) + a \quad (\text{add constant to preserve DC}) \quad (20)$$

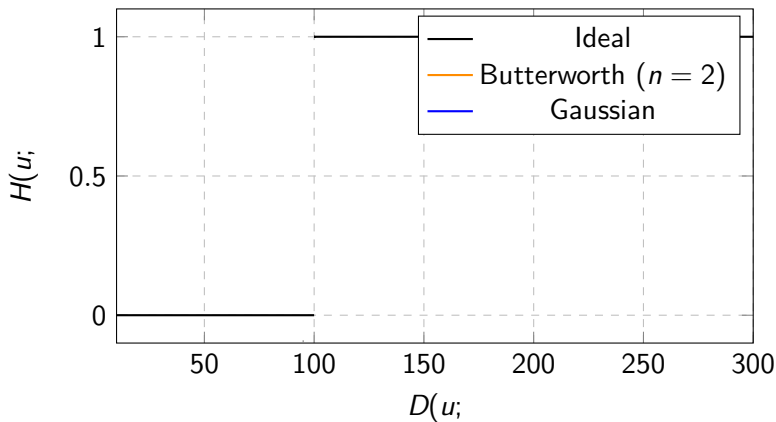
- **Purpose:** Restore original image structure while emphasizing edges

Comparison: IHPF vs BHPF vs GHPF vs Laplacian

Aspect	IHPF	BHPF	GHPF	Laplacian
Formula	$\begin{cases} 0 & D \leq D_0 \\ 1 & D > D_0 \end{cases}$	$\frac{1}{1+[D_0/D]^{2n}}$	$1 - e^{-D^2/(2D_0^2)}$	$-D^2$
At Origin	0	≈ 0	0	0
Cutoff	Sharp	Smooth	Smooth	Parabolic
Ringing	Severe	Moderate	None	None
Artifacts	Halo	Minimal	Minimal	Sensitive to noise
Parameters	1	2	1	0
Practical Use	Theory only	Common	Preferred	With caution
Edge Detection	Poor	Good	Excellent	Excellent

Frequency Response Comparison

All High-Pass Filters Compared



Gaussian provides smoothest response; Ideal has discontinuity

Edge Detection and Enhancement

- **High-Pass Filtering for Edge Detection:**

- Removes background (low frequencies)
- Enhances boundaries and transitions
- Reveals fine details
- Ideal for feature extraction

- **Laplacian Best for Edges:**

- Second-order derivative captures rapid changes
- Detects edges of all orientations
- Very sensitive and precise
- Often combined with original image

- **GHPF Best for Quality:**

- Smooth response minimizes artifacts
- No ringing effects
- Natural-looking enhancement
- Balances edge detection with image quality

Image Sharpening Applications

- **Unsharp Masking:**

$$g(x, y) = f(x, y) + c \cdot [f(x, y) - f_{\text{blur}}(x, y)] \quad (21)$$

In frequency domain:

$$G(u, v) = F(u, v) + c \cdot [F(u, v) - H_{\text{LPF}}(u, v)F(u, v)] \quad (22)$$

- **High-Boost Filtering:**

$$H_{\text{boost}}(u, v) = a + b \cdot H_{\text{HPF}}(u, v) \quad (23)$$

where a restores image brightness

- **Common Use Cases:**

- Enhance blurry images
- Improve clarity of documents
- Emphasize texture details
- Preprocessing for analysis

Feature Extraction and Analysis

- **Feature Extraction:**

- Isolate edges and corners
- Detect fine structures
- Separate details from background
- Extract structural information

- **Medical Imaging:**

- Enhance details in X-rays and CT scans
- Emphasize anatomical structures
- Improve diagnostic information
- Combined with Laplacian for precision

- **Material Inspection:**

- Detect surface defects
- Identify cracks and flaws
- Quality control applications
- Texture analysis

Implementation: Practical Steps

- **Step 1: Prepare Image**

- Pad to avoid circular convolution
- Use power of 2 for FFT efficiency

- **Step 2: Compute 2-D DFT**

$$F(u, v) = \text{FFT2D}(f(x, y)) \quad (24)$$

- **Step 3: Center and Apply Filter**

$$G_c(u, v) = H(u, v) \cdot F_c(u, v) \quad (25)$$

- **Step 4: Inverse Transform**

$$g(x, y) = \text{Real}(\text{IFFT2D}(G_c(u, v))) \quad (26)$$

- **Step 5: Post-processing**

- Scale to valid range (if needed)
- Combine with original for visualization

Parameter Selection Guidelines

- **Cutoff Frequency D_0 (for BHPF and GHPF):**
 - **Small D_0 ($\approx 10 - 30$):** Heavy edge emphasis, aggressive enhancement
 - **Medium D_0 ($\approx 50 - 100$):** Balanced edge enhancement
 - **Large D_0 ($\approx 100 - 200$):** Subtle enhancement, detail preservation
 - **Typical:** 5% to 10% of image dimension
- **Butterworth Order n :**
 - $n = 1$: Smooth response, gentle enhancement
 - $n = 2$: Good balance (recommended)
 - $n = 4$: Sharper cutoff
- **Selection Strategy:**
 - Use Gaussian for best results
 - Laplacian for precise edge detection
 - Butterworth for adjustable control

Algorithm: High-Pass Filtering

[H] Frequency Domain High-Pass Filtering **Input:** Image $f(x, y)$, Filter type, Parameters
Output: Filtered image $g(x, y)$ 1. Pad image to $(2M) \times (2N)$ (or power of 2) 2.

Compute 2-D DFT: $F(u, v) \leftarrow \text{FFT2D}(f)$ 3. Center spectrum:

$F_c(u, v) \leftarrow F(u, v) \cdot (-1)^{u+v}$ 4. Create distance matrix: $D[i, j] \leftarrow \sqrt{(i-M)^2 + (j-N)^2}$
5. **Compute filter** $H[i, j]$: **if** Filter = IHPF: $H[i, j] \leftarrow (D[i, j] > D_0) ? 1 : 0$ **elseif**

Filter = BHPF (order n): $H[i, j] \leftarrow 1 / (1 + [D_0 / D[i, j]]^{2n})$ **elseif** Filter = GHPF:
 $H[i, j] \leftarrow 1 - e^{-D[i, j]^2 / (2D_0^2)}$ **elseif** Filter = Laplacian: $H[i, j] \leftarrow -D[i, j]^2$ 6. Apply

filter: $G_c(u, v) \leftarrow H(u, v) \times F_c(u, v)$ 7. Inverse transform: $g(x, y) \leftarrow \text{Real}(\text{IFFT2D}(G_c))$
8. **Return** $g(x, y)$

High-Boost Filtering

- **Concept:** Combine high-pass effect with original image
- **Mathematical Definition:**

$$H_{HB}(u, v) = k + H_{HPF}(u, v) \quad (27)$$

where k is boost factor ($k \geq 1$)

- **Spatial Domain Equivalent:**

$$g(x, y) = f(x, y) + c \cdot [f(x, y) - f_{\text{blur}}(x, y)] \quad (28)$$

- **Advantages:**
 - Preserves original image information
 - Enhanced edges without darkness
 - Better visual quality
 - More intuitive results

Unsharp Masking in Frequency Domain

- **Classical Unsharp Mask:**

$$g(x, y) = f(x, y) + \lambda[f(x, y) - f_{\text{blur}}(x, y)] \quad (29)$$

where λ is sharpening strength

- **In Frequency Domain:**

$$H_{\text{USM}}(u, v) = 1 + \lambda[1 - H_{\text{GLPF}}(u, v)] \quad (30)$$

- **Effect:**

- Enhances edges significantly
- Preserves lower frequencies
- Produces crisp, sharp images
- Widely used in image enhancement

- **Parameters:**

- D_0 : Controls blur amount
- λ : Controls enhancement strength

Summary Table: High-Pass Filters

Filter	Key Equation	Best For	Issues
Ideal	$H = \begin{cases} 0 & D \leq D_0 \\ 1 & D > D_0 \end{cases}$	Theory	Severe ringing
Butterworth	$H = \frac{1}{1 + [D_0/D]^{2n}}$	Control	Noise sensitive
Gaussian	$H = 1 - e^{-D^2/(2D_0^2)}$	Practice	Gentle rolloff
Laplacian	$H = -D^2$	Edge detect	Very noisy

Property	IHPF	BHPF	GHPF	Laplacian
Ringing	Severe	Moderate	None	None
Implementation	Simple	Medium	Simple	Simple
Parameters	1	2	1	0
Practical Use	No	Yes	Preferred	With caution
Quality	Poor	Good	Excellent	Excellent
Noise	Amplified	Very amplified	Amplified	Very amplified

Textbook and Reference Materials

Recommended Reading

- **Textbook:**

- Rafael C. Gonzalez and Richard E. Woods, *Digital Image Processing*, Pearson Edition, Latest Edition.

- **Reference Books:**

- I. Pitas, *Digital Image Processing Algorithms*, Prentice Hall, Latest Edition.
- A. K. Jain, *Fundamentals of Digital Image Processing*, Prentice Hall of India Pvt. Ltd., Latest Edition.
- K. Castleman, *Digital Image Processing*, Prentice Hall of India Pvt. Ltd., Latest Edition.

- **Related Topics:**

- Band-pass filters (combine low-pass and high-pass)
- Notch filters (frequency-selective attenuation)
- Edge detection and feature extraction

Key Takeaways - Sharpening Filters

- **Ideal High-Pass Filter:** Sharp cutoff but severe ringing artifacts
- **Butterworth High-Pass Filter:** Smooth transition; good control with order
- **Gaussian High-Pass Filter:** Best practical choice; smooth, artifact-free
- **Laplacian Filter:** Excellent for edge detection; very sensitive to noise
- **High-Boost Filtering:** Combines sharpening with original for better results
- **Unsharp Masking:** Classical technique; effective for quality enhancement

Practical Recommendations:

Use Gaussian for general sharpening;
Laplacian for edge detection (with preprocessing);
High-boost for best visual quality